



The value-creating potential of AI: A multi-dimensional analysis of effects and mechanisms

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ABSTRACT

The integration of artificial intelligence in corporate operations has emerged as a transformative force in business landscapes, yet existing research provides limited insights into the mechanisms through which AI creates enterprise value. Drawing on a comprehensive dataset of Chinese listed companies spanning 2010–2022, this study investigates how AI adoption influences enterprise value through the mediating role of data assets and the moderating effect of managerial capability. Our analysis reveals that AI adoption significantly enhances enterprise value both directly and indirectly through data assets, demonstrating substantial mediation effects. The findings establish that managerial capability positively moderates this relationship, indicating that firms with stronger management teams extract greater value from AI investments. Through extensive robustness checks and endogeneity tests, we establish the causal nature of these relationships. Three heterogeneity tests reveal boundary conditions: the valuation premium is stronger for state-owned enterprises, disappears in heavily polluting industries unless AI is part of a credible green transition, and is evident only in non-asset-intensive firms where resource reconfiguration is agile. By integrating resource-based, dynamic-capability, and stakeholder perspectives, the study clarifies when AI becomes a genuine source of competitive advantage. The results inform executives that data governance, leadership development, and asset or environmental restructuring must accompany AI budgets, and they guide policymakers toward incentive designs that couple digital adoption with sustainability and capital-efficiency objectives.

1. Introduction

The digital revolution is no longer a distant horizon but an immediate operational reality reshaping how firms generate, appropriate, and sustain value. Artificial intelligence (AI) now occupies the epicentre of this upheaval, serving simultaneously as an innovation enabler, a process optimiser, and a strategic decision partner across virtually every sector, from heavy manufacturing and logistics to financial services and creative industries (Bloom, Prettnner, Saadaoui, & Veruete, 2025; Bonaparte, 2024; Xin, 2025; Yang, 2025). Global spending on AI-centric solutions eclipsed USD 150 billion in 2024 and is forecast to double within four years, signalling that boardrooms increasingly view machine learning, natural-language processing, and computer vision not as peripheral IT utilities but as core capabilities essential for long-term competitiveness (Durongkadej, Hu, & Wang, 2024; Kurup & Gupta,

2022; Zhou, Li, Wang, Li, & Zhou, 2025). Yet this widespread enthusiasm is tempered by persistent execution gaps: surveys reveal that only a minority of deployments move beyond experimental pilots into scaled, revenue-impacting applications, while failure rates for enterprise AI projects still exceed 50 % (Han, Li, & Jin, 2025; Lee, Kim, & Ivan, 2023). Early adopters such as leading e-commerce and fintech platforms demonstrate that data-driven algorithms can compress cycle times, personalise customer journeys, and unlock entirely new revenue streams, but many traditional incumbents struggle to translate proof-of-concept successes into firm-wide performance gains (Yin, Yin, & Wen, 2025). These mixed outcomes suggest that the value-creating capacity of AI depends on more than technological sophistication alone; it hinges on complementary intangible resources, effective governance structures, and context-specific managerial capabilities—factors that current empirical research has only partially explored.

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